Case Study Report

Design and Creative Technologies

Torrens University, Australia

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Designing a Big Data Pipeline and Data Lake for Big Retail

1. Executive Summary

Big Retail, an Adelaide online retailer with more than 100,000 monthly website visitors, has lost sales and conversion despite holding a cost advantage (Torrens University Australia, 2024). The root cause is structural rather than commercial: a static website serves identical content to every visitor, while customer, sales, and marketing data sit in separate departmental databases that cannot be combined to personalize the experience. This report designs the data pipeline that must precede any analytics work. It identifies the internal and external data sources aligned to Big Retail's data-driven goals (targeted campaigns, a recommender system, and product association), articulates the integration challenges that must be resolved before storage, and specifies a governed lakehouse-style data lake on AWS — with open-source equivalents — that stores structured, semi-structured, and unstructured data and serves it efficiently to both managers and data scientists. Key technical terms used throughout are defined in Appendix A.

2. Data strategy and the six V’s for Big Retail

A data-driven strategy must start from the business problem, not the technology: data projects fail when tools are chosen before the organization is clear on *why* the work matters (Marr, 2021). For Big Retail the “why” is conversion recovery through personalization. Evaluating the six V’s shows which characteristics actually drive the design rather than treating them as a checklist (Rutherford, 2017).

- **Volume** – moderate but sufficient: 100,000+ monthly visitors plus transactions, emails, and catalogue data give enough behavioral signal for pattern detection. Volume justifies object storage and distributed (Spark) processing but is not the binding constraint.
- **Velocity** – selectively important: only clickstream and cart events are continuous; sales, CRM, and catalogue data are batch. The design therefore needs streaming **and** batch ingestion, not streaming everywhere.
- **Variety** – the defining challenge: relational transactions, JSON clickstream, API feeds, and free-text reviews must coexist, which is precisely what a data lake (not a warehouse) is built to hold.
- **Veracity** – the costliest risk: duplicate customers across sales and marketing systems, guest-checkout gaps, and bot traffic threaten every downstream model, so data quality is treated as a first-class pipeline stage.
- **Valence** – the value unlock: the recommender and product-association use cases depend on c*onnecting* customers to sessions, orders, products, and campaigns. Raising valence through identity resolution is where most analytical value is created.
- **Value** – the anchor: every source and component below is justified only by its contribution to campaigns, recommendations, association rules, or conversion reporting.

In short, Variety, Veracity, and Valence – not raw Volume – dominate Big Retail's design. Appendix D maps each V to its design response.

3. Potential data sources

The intended analytics cannot run on transactions alone; recommenders and targeted campaigns need behavioral, descriptive, and contextual data, and product association needs reliable order-line data linked to categories and campaign timing. The full inventory (Appendix E, Table E1) spans **internal and external** sources and all three structure types, with fields and formats taken as reasonable assumptions for an online retailer of this size.

These sources fall into four characteristic groups. **Transactional and account data** (sales DB, customer accounts) are highly structured, low-velocity, and high-veracity when controlled, but PII-sensitive and prone to duplication across the sales and marketing teams – they anchor revenue analysis and segmentation. **Behavioral data** (clickstream, search, cart events) is the highest-velocity and highest-volume source; it arrives as semi-structured Json, carries bot noise, and is the raw signal for funnel analysis and the recommender, yet it depends on the IT team being able to emit or export events. **Descriptive data** (product catalogue, promotions) is a mix of structured attributes and semi-structured media that changes frequently and must be treated as slowly changing dimension, so recommendations stay stock aware. **Contextual and unstructured data** (reviews, complaint text, weather, demographics, competitor signals) adds the variety that lifts personalization beyond transactions but brings the weakest veracity and the clearest legal constraints, so it is enriched cautiously.

Together, these cover **structured** (transactions, accounts, catalogue), **semi-structured** (clickstream, CRM events, weather/API payloads), and **unstructured** (review text, complaint notes, competitor/social signals) data – the full spread the recommender, campaign, and association use cases require, and the spread a warehouse alone could not hold.

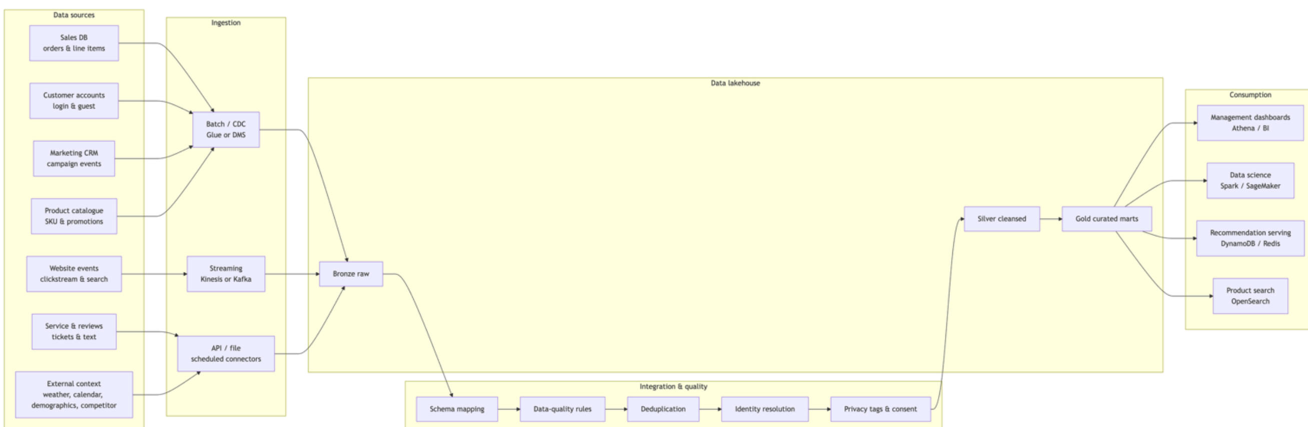
4. Data integration challenges and resolution strategy

The hard problem is not storage; it is joining fragmented customer, product, campaign, and event data without fabricating false identities or exposing personal information. Integration is therefore treated as a *governed* stage, not a one-off load – the data-preparation work that, more than the modelling itself, determines the quality and credibility of every downstream result (EMC Education Services, 2015). Schema alignment and duplicate resolution – the two challenges the rubric names explicitly – head the list.

The two challenges the rubric names deserver fuller treatment. **Schema alignment** fails silently rather than loudly: the sales DB may call a key *cust_id* while marketing calls it *customer_ref* and the catalogue keys products by an internal *item_code* that the promotions feed does not share. Resolving this means agreeing canonical entities (*customer, product, order, campaign, event*), publishing them in a versioned schema registry, and validating each incoming feed against a data contract so a renamed or retyped field is rejected at ingestion rather than corrupting a join downstream. **Duplicate customers** are the costlier problem because the case explicitly allows the same person to exist in both the sales nad marketing databases and to move between guest checkout, a new account and a an existing login. Resolution is staged: a deterministic pass matches on hashed email or account ID first, and only unmatched records fall through a cautious fuzzy pass on name, address, and phone, every match carrying a confidence score a full source lineage so a merge can be audited or reversed.

Beyond those two, two resolutions carry real trade-offs worth stating. **Identity matching** must favor precision over recall: aggressive fuzzy matching merges distinct customers, corrupting both recommendations and privacy boundaries, so low-confidence matches are flagged rather than merged. **Batch-vs-streaming** is not a free choice either – streaming everything adds cost and operational complexity for data (sales, catalogue) that changes slowly, so streaming is reserved for clickstream where latency genuinely matters. The end-to-end flow is shown in **Figure 1**; Appendix B (Figure B1) expands the integration stage into its step-by-step validation and resolution sequence. Data quality (bot filtering, deduplication, malformed-field handling) and external-feed reliability are handled as further governed steps; the full challenge register, naming each issue with a resolution step and output artefact, is given in **Appendix G (Table G1)**.

Figure 1– End-to-end Big Retail data pipeline and lakehouse (primary schematic)



5. Data lake design: storage, retrieval, security

5.1. Architecture

The recommended design is a **lakehouse on cloud object storage**, because a data lake can hold structured and unstructured data at scale and expose it for dashboards, batch processing, and machine learning (Amazon Web Services, n.d.-b). AWS is proposed as the primary stack - Big Retail's team already meets the CCF501 cloud prerequisite - with open-source equivalents (full mapping in Appendix F, Table F1) so the design is portable and not locked in.

The AWS-vs-open-source choice is a genuine trade-off: managed services cut operational burden and speed delivery but add cost and vendor lock-in, whereas the open-source stack lowers licensing cost at the price of in-house operations. Open table formats (Iceberg/Delta) and S3-compatible storage keep migration feasible either way.

5.2. Lake zones

Data moves through three governed zones - **Bronze** (immutable raw data: DB extracts, clickstream JSON, review text), **Silver** (standardized, deduplicated, privacy-tagged Parquet), and **Gold** (business-ready marts and features in Iceberg/Delta) - plus a **serving** tier for low-latency outputs. This medallion layering is what lets the lake hold every structure type while still serving fast queries; Appendix H (Table H1) details each zone's purpose, examples, and format.

5.3.Retrieval

Retrieval is split to satisfy both audiences and both rubric retrieval requirements. Efficient search for managers and data scientists uses SQL engines (Athena, Redshift Spectrum, or Trino) over Parquet/Iceberg tables partitioned by date, source, and category (Amazon Web Services, n.d.-a). Low-latency retrieval for the live site is served from precomputed gold outputs pushed into DynamoDB/Redis (e.g. customer_id →

top_product_ids) and OpenSearch for product/review search - so the website never queries raw lake files during checkout, while the lake stays the governed source of truth.

5.4. Security and privacy

Because accounts, campaign data, and behavioral logs contain personal information, collection and storage follow data-protection best practice aligned to the Australian Privacy Principles (Office of the Australian Information Commissioner, n.d.): minimize to only the fields each use case needs; encrypt in transit and at rest; enforce role-based access (managers see aggregates, scientists see de-identified analytical sets, engineers administer pipelines); hash/mask direct identifiers and separate identity tables from behavior tables; store consent state as first-class fields; and catalogue every dataset with owner, lineage, and retention period for auditability.

6. Assumptions, limitations, and conclusion

The design rests on assumptions to confirm with Big Retail: that the IT team can export or emit clickstream events; that guests can be linked via hashed email, device/session ID, or later sign-up; that sales and marketing share overlapping (if inconsistent) customer fields; and that external competitor/social data is collected only through lawful APIs or compliant providers. Its limitations are honest ones – identity resolution will never be perfect for guest traffic, external feeds carry variable veracity, and the lakehouse adds operational cost that only pays off once the recommender and campaign use cases are live. Subject to those caveats, the proposed pipeline ingests Big Retail's fragmented internal and external sources, resolves schema and duplicate-identity challenges before storage, and lands data in a governed lakehouse that stores structured, semi-structured, and unstructured data while serving both analytical and low-latency retrieval. Crucially, every component traces back to business value: targeted campaigns, personalized recommendations, product-association insight, and recovered conversion – Appendix C (figure C1) maps each curated dataset to the business outcome it serves.

7. Appendices

7.1. Appendix A – Glossary of Terms

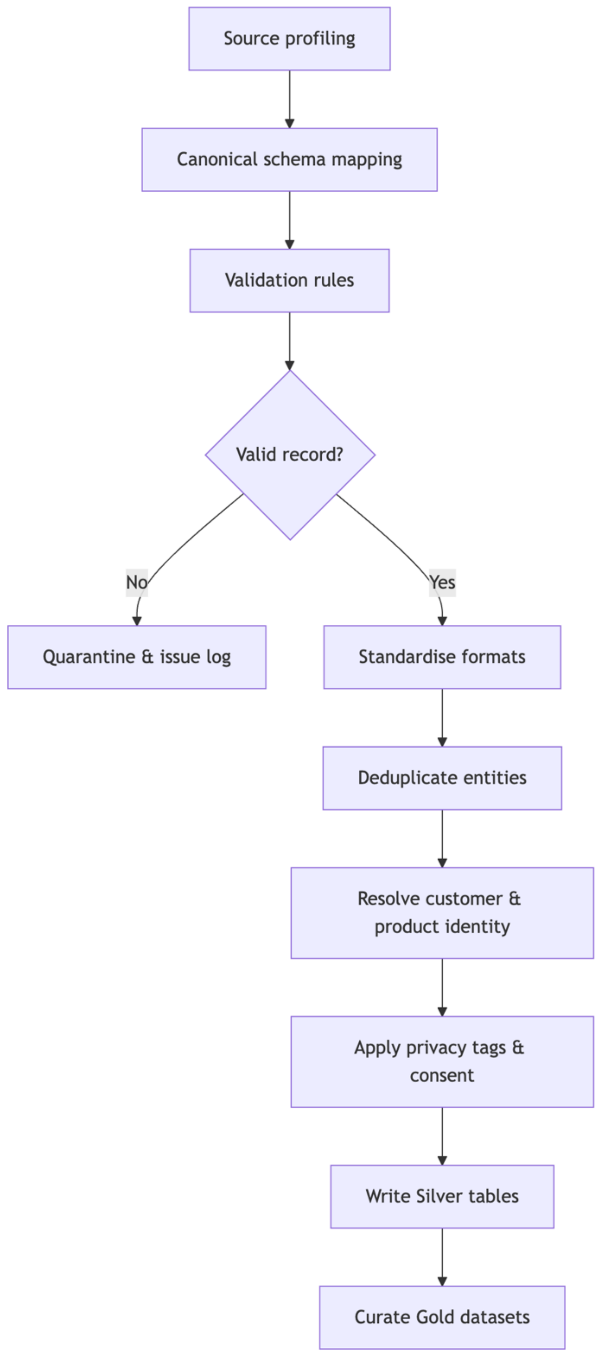
Table 5 – Term and Definition support table.

Term	Definition
Data lake	A store-all repository that holds raw data of any structure until it is needed, applying structure on read.
Lakehouse	An architecture that adds warehouse-style ACID tables and SQL querying on top of a data lake's low-cost object storage.
Schema-on-read	Structure is applied when data is queried, not when it is stored (contrast schema-on-write in a warehouse).
Medallion (Bronze / Silver / Gold)	Layered lake zones: raw landing (Bronze), cleansed and standardized (Silver), business-ready marts (Gold).
ELT (land-then-validate)	Load raw data first, then transform and validate it inside the lake (contrast ETL).
CDC (Change Data Capture)	Ingesting only the records that change since the last load, instead of re-copying whole tables.
Identity resolution	Linking records that refer to the same customer across sources, despite inconsistent or missing IDs.
Event-time watermark	A threshold defining how late a streaming event may arrive before its processing window closes.
PII	Personally identifiable information (names, emails, addresses) requiring privacy controls.
OAIC APPs	The Australian Privacy Principles, the national baseline for handling personal information.
Lineage	A record of where a dataset came from and how it was transformed, used for audit and trust
Parquet / Iceberg / Delta	Columnar file and ACID table formats used for efficient analytical storage in the lake

7.2.Appendix B – Integration workflow

Figure B1 expands the "Integration & quality" stage of Figure 1 into its step-by-step validation and entity-resolution sequence, including the quarantine path for records that fail validation.

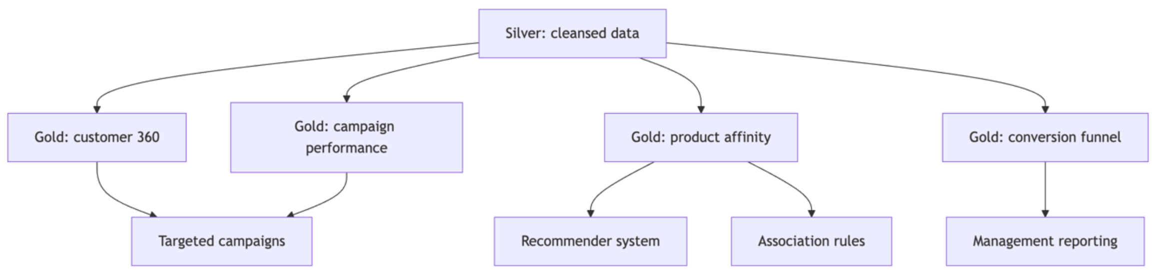
Figure B1 – Integration workflow (source profiling to curated)



7.3. Appendix C – Curated datasets and business value

Figure C1 traces how curated (Gold) datasets map to Big Retail's target business outcomes, showing the line of sight from cleansed data to campaigns, recommendations, association rules, and management reporting.

Figure C1 – Curated outputs to business value



7.4. Appendix D – The six V's mapped to Big Retail

This table cross-references the evaluation in §2: each V is mapped to its Big Retail interpretation, the concrete design response, and the severity that drives the design.

Table 6 – Six V's mapping to Big Retail

V	What it means for Big Retail	Design response	Severity here
Volume	100,000+ monthly visitors plus transactions, emails, catalogue, and events	S3 object storage and distributed Spark processing	Moderate – not the binding constraint
Velocity	Clickstream and cart events are continuous; sales, CRM, and catalogue are batch	Streaming (Kinesis/Kafka) for events plus batch/CDC for databases	Selective
Variety	Relational, JSON, API, and free-text data must coexist	Lakehouse zones store structured, semi-structured, and unstructured data	Primary

Veracity	Duplicate customers, guest-checkout gaps, and bot traffic	Data-quality rules, deduplication, identity resolution, and quarantine	Primary
Valence	Customers link to sessions, orders, products, and campaigns	Identity graph plus customer and product master tables	Primary (value unlock)
Value	Every component must serve a business outcome	Gold marts and serving features tied to campaigns, recommendations, association and conversion	Anchor

7.5. Appendix E – Data-source inventory

The full per-source inventory referenced in §3: internal and external sources across all three structure types, with assumed fields, formats, and the business value each serves.

Table E1 – Big Retail data-source inventory

Source	Int/Ext	Structure	Assumed fields & format	Business value
Sales transaction DB	Internal	Structured	order_id, customer_id, guest_id, sku, qty, price, discount, payment_status, order_ts; relational / csv	Association rules, revenue, conversion
Customer account DB	Internal	Structured	customer_id, name, email, address, postcode, prefs, created_at; relational (PII-sensitive)	Segmentation, personalization, retention
Marketing / email CRM	Internal	Structured + semi	campaign_id, segment, send_ts, open/click/bounce/unsub events; CSV / JSON / API	Targeted campaigns, attribution
Website clickstream & search logs	Internal	Semi-structured	session_id, visitor_id, url, referrer, query, product_view, cart_event, ts, device, geo; JSON events	Funnel analysis, recommender signals

Product catalogue & promotions	Internal	Structured + semi	sku, category, brand, attributes, stock, price, images, promo, margin; relational / CSV	Similarity, offer ranking, stock-aware recs
Customer service, returns & reviews	Internal	Structured + **unstructured**	ticket_id, reason_code, free-text complaint, return_reason, rating, review_text; DB + text	Pain points, sentiment, product feedback
Location / demographic / calendar	External	Structured	postcode population, income bands, holidays, local events; CSV / API	Contextual campaign segmentation
Weather	External	Structured + semi	date, postcode, temperature, rainfall, alerts; API / JSON	Promotion timing, demand modelling
Competitor pricing public signals	External	Semi + **unstructured**	competitor price, title, availability, public reviews / social mentions; API / lawful scrape	Pricing intelligence, demand signals

7.6Appendix F – Storage and retrieval stack

The full storage and retrieval stack referenced in §5.1, mapping each AWS service to an open-source equivalent so the design stays portable.

Table F1 –Storage and retrieval stack (AWS primary, with open-source equivalents).

Layer	AWS (primary)	Open-source equivalent	Role
Batch ingestion	Glue / DMS	Airbyte, Debezium	Import sales, CRM, catalogue
Streaming ingestion	Kinesis	Apache Kafka	Clickstream, cart, checkout events
Raw storage	Amazon S3	MinIO / HDFS	Durable object storage, all raw files
Processing	Glue / EMR Spark	Apache Spark	Clean, join, aggregate, feature build
Table format	Iceberg / Delta	Iceberg / Delta	ACID tables, schema evolution, time travel

Catalogue	Glue Data Catalog	Hive Metastore, Open Metadata	Searchable schemas, ownership, lineage
Governance	Lake Formation	Apache Ranger	Fine-grained access, audit
Analytical query	Athena / Redshift Spectrum	Trino / Presto	SQL for dashboards and analysts
Low-latency serving	DynamoDB / OpenSearch	Cassandra / Redis / OpenSearch	Fast recommendation & product lookup

7.7 Appendix G – Integration challenge register

The full set of integration challenges referenced in §4: schema alignment and duplicate resolution first (the two the rubric names), then identity, timing, quality, privacy, and external-source issues, each with a resolution step and the artefact it produces.

Table G1 – Integration challenges and resolution steps.

Challenge	Why it appears at Big Retail	Resolution step	Output artefact
Schema alignment	Sales, marketing, web, and catalogue systems use different field names, types and IDs	Define canonical *customer / product / order / campaign / event* schemas; validate incoming data against data contracts	Versioned schema registry + canonical model
Duplicate customers	Same customer in sales **and** marketing DBs; guests later create accounts	Deterministic match on hashed email / account ID, then *cautious* fuzzy match on name/address/phone with confidence scores	Master customer table with source lineage

Guest-checkout identity	Guest orders may not map to web session or later accounts	Store privacy-safe guest_id, session ID, hashed email, and linkage timestamps	Identity graph with transparent lineage
Batch vs streaming mismatch	DB/CRM are batch; clickstream is continuous	Batch/CDC for databases, streaming for events; watermark and reconcile late events on event-time	Bronze landing zone + event-time rules
Data quality	Address formats, bot traffic, duplicate events, missing profiles	Validation rules, standardize timestamps to UTC/AEST, dedupe event IDs, filter bots, quarantine bad records	Data-quality scorecard + quarantine table
Privacy & consent	Accounts, campaigns, and logs hold PII	Minimization, encryption, role-based access, consent flags, masking, retention, audit logging (OAIC APPs)	Privacy control matrix
External-source reliability	Weather/competitor/review feeds refresh unevenly	Record source, extraction time, license, reliability score, refresh cadence	External source register

7.8 Appendix H – Lakehouse zones

The medallion zones referenced in §5.2, from raw landing to curated marts plus a serving tier, showing how each structure type is stored.

Table H – Lakehouse zones (Raw to curated, plus a serving tier)

Zone	Purpose	Examples	Format
Bronze (raw)	Immutable source data for audit/replay	DB extracts, raw clickstream, JSON, review text	CSV, JSON, Avro, text

Silver (cleansed)	Standardized, deduplicated, privacy-tagged	Master customer/product, cleaned orders, standardized events	Partitioned Parquet
Gold (curated)	Business-ready marts and features	Customer 360, product-affinity matrix, campaign performance	Iceberg / Delta
Serving	Fast retrieval for live use	Top-N recommendations, product search index	Key-value / search index

End of Appendix Section

Statement of Acknowledgment

I acknowledge that I have used the following AI tool(s) in the creation of this report:

- OpenAI ChatGPT (Codex-5.5)
- Anthropic Claude Sonnet 4.8

Both have been used to assist with outlining, refining structure, improving clarity of academic language, and supporting APA 7th referencing conventions.

Prompt examples:

1. *"I'm writing a 1,500-word case study report on Funk et al. (2020) - a framework for applying NLP in digital health interventions. Here is the assessment rubric [pasted]. Help me write a 200–250 word Introduction section that covers the problem significance and outlines the report structure. Do not add content I haven't asked for."*
2. *"Here is a numbered list of seven DHITA feature engineering categories with counts and methods. Convert this into a clean markdown table with four columns: Feature Type, Count, Method/Tool, Key Reference. Preserve all existing data exactly."*
3. *"My §3.3 ethics section is one dense 90-word paragraph. Rewrite it as a one-sentence framing statement followed by a four-row table. Rows: Privacy & consent, Bias & fairness, Explainability, Clinical safety - each with an Issue column and a Risk Level (High / Medium / Critical). Keep total word count similar to the original."*

I confirm that the use of the AI tool has been in accordance with the Torrens University Australia Academic Integrity Policy and TUA, Think and MDS's Position Paper on the Use of AI. I confirm that the final output is authored by me and represents my own critical thinking, analysis, and synthesis of sources. I take full responsibility for the final content of this report.

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